

Water Sustainability Practices in Dialysis

Executive summary

The strongest *implemented* water-sustainability practices in dialysis today are not the most futuristic ones. They are the pragmatic measures applied around the dialysis water plant: reusing reverse osmosis reject water for non-patient-contact purposes, upgrading or optimizing reverse osmosis systems, avoiding redundant disinfection cycles, and recovering heat from warm effluent. These measures are already producing quantified results in routine care. UK and Australian case studies report reject-water reuse ranging from about **1,000 L/week** in a small rural unit to **13,140 m³/year** in a children's hospital; direct cost savings range from **£7,500/year** at Canterbury to **more than £15,000/year** at Bristol, with some projects showing payback in a few years rather than decades. Operational optimization also matters: a multicentre Spanish study found that centres operating daily used materially less water and energy per dialysis session than centres opening only three times weekly, indicating that scheduling and asset utilization are themselves sustainability levers. ¹

On the technology side, the best current evidence favors “**reduce and reuse**” rather than full “**recycle**” of patient-contact effluent. Newer water plants and machine configurations can materially reduce resource intensity: a recent Australian comparison reported **357 L** of RO water and **3.1 kWh** of RO power per treatment with a modern RO system versus **548 L** and **7.2 kWh** with an older one. Manufacturers such as B. Braun and Fresenius Medical Care advertise **up to 85% RO water recovery** on newer systems, and Fresenius reports wider deployment of heat disinfection and water metering in water-stressed clinics. Heat-recovery interventions are also maturing: retrofitted dialysis-machine heat exchangers in NHS services saved **0.5–0.86 kWh per treatment**, while a 2026 simulation/experimental validation for a four-station clinic projected **>15 kWh/session** and **~63,200 kWh/year** recovered. ²

By contrast, **closed-loop spent-dialysate regeneration**, **forward osmosis**, and **portable sorbent hemodialysis** remain promising but comparatively immature. Sorbent systems can dramatically cut water demand: the legacy REDY system required about **6 L** of untreated water per dialysis, and NextKidney's NeoKidney reports **4.5–5 L** of dialysate per session with no separate home water treatment system, but the company also states that the device is **not yet for sale** and that regulatory approvals are pending. The clinical safety evidence retrievable in this review is still limited: an AWAK peritoneal dialysis pilot reported **no serious adverse events** in 15 patients but did report frequent abdominal discomfort, underscoring that regenerative approaches are not yet “drop-in” replacements for conventional dialysis infrastructure. ³

The policy picture is mixed. Water-quality regulation for dialysis is already strict and well established; sustainability regulation and incentives are newer and more fragmented. Current rules mainly act as a **boundary condition**: they protect patient safety, but they also make any reuse that re-enters patient-contact pathways far more complex than reusing reject water for toilets, boilers, irrigation, or laundry. The most scalable implementation pattern emerging from the evidence is therefore: **measure first, optimize existing plants and schedules second, reuse non-patient-contact streams third, and reserve closed-loop patient-contact systems for controlled pilots until the clinical, regulatory, and economic evidence base is stronger**. ⁴

Scope and evidence base

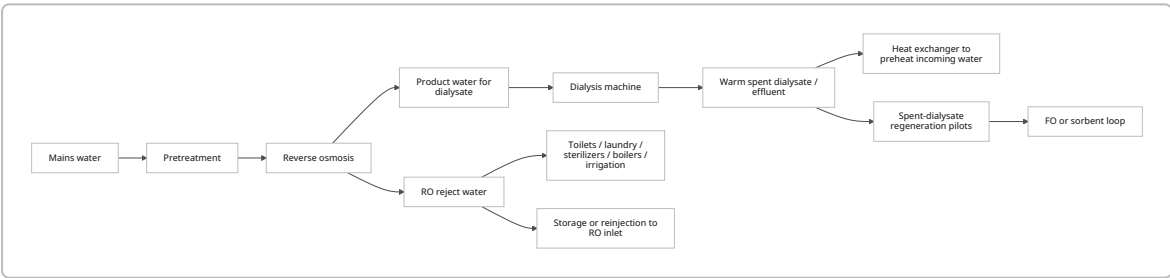
This report prioritizes peer-reviewed papers, official hospital and dialysis-service case studies, manufacturer technical documentation, professional-society guidance, and government or health-system documents. The primary evidence retrieved clusters around several source types: multicentre observational studies from Spain and Australia; hospital implementation case studies from the UK and Australia; official technical or sustainability material from Fresenius Medical Care, B. Braun, DaVita, and NextKidney; and guidance from WHO, the NSW Agency for Clinical Innovation, ANZSN, ERA/SFNDT, KDIGO, and ISN GREEN-K. ⁵

A key limitation of the literature is its asymmetry. Quantified evidence is comparatively strong for **RO reject reuse, water-plant upgrades, disinfection scheduling, and heat recovery**. It is weaker for **patient-level clinical outcomes** directly attributable to sustainability interventions, and much weaker still for **closed-loop spent-dialysate recycling** and **portable sorbent hemodialysis**. Where sources did not specify a result, this report records it as **unspecified** rather than inferring it. ⁶

Assumption used in this report	Treatment of the issue
Scalability is rated qualitatively	It reflects technical/regulatory replicability, not guaranteed business-case attractiveness in every water tariff or building context.
Manufacturer performance claims are not treated as field-validated by default	They are labelled as manufacturer-reported unless an independent implementation study was also retrieved.
“Patient grouping” is discussed as an operational inference	It is inferred from utilization and scheduling evidence, not from a standalone trial explicitly randomizing grouping strategies.
Corporate sustainability metrics may include extrapolation	Reported exactly as disclosed by the company, with method caveats where stated by the source.

Technical practices

The dialysis water chain has several intervention points, and the evidence suggests that the most robust gains come from intervening at the **RO plant boundary** and the **effluent/heat boundary**, not from modifying the patient-contact dialysate loop itself. The most mature practices are therefore: increasing RO efficiency, reusing reject water, right-sizing and scheduling disinfection, and recovering thermal energy from wastes. More ambitious approaches—direct spent-dialysate purification, forward osmosis regeneration, and sorbent recirculation—remain largely in pilot, prototype, or early clinical stages. ⁷



This schematic reflects intervention points described in hospital case studies, operational handbooks, and manufacturer materials. The practical divide is important: **RO reject water** has never contacted the patient and is usually treated as a fit-for-purpose reuse stream after local assessment; **spent dialysate** is a much more demanding technical and regulatory challenge, even when its resource content is attractive. ⁸

Technology comparison

Practice	What is being done	Measured or reported effect	Safety and evidence status
RO reject-water reuse	Recover reject water from the RO plant for toilets, laundry, sterilizers/steam, cooling towers, irrigation, or other non-patient-contact uses	Real-world projects report 800 L/h recovered at Canterbury; 1.7 ML/year at Bendigo; 1.1 ML/year estimated at Barwon; 1.5–2 ML/year at Queen Elizabeth Hospital; 13,140 m³/year at Bristol. ⁹	Strongest real-world evidence base. Safety case is strongest when reuse remains outside patient-contact pathways; site-level water-quality and plumbing controls remain necessary. ¹⁰
RO optimization and high-recovery systems	Replace legacy plants, optimize recovery settings, reduce unnecessary regeneration/disinfection, and specify newer RO designs	A recent Australian comparison reported 357 L RO water and 3.1 kWh RO power per treatment with a modern system versus 548 L and 7.2 kWh with an older system. B. Braun and Fresenius report designs with up to 85% water recovery . ¹¹	Good evidence for resource savings; mixed evidence on site-specific payback because costs depend on plant age, tariffs, and retrofit complexity. Manufacturer recovery claims are not the same thing as independently audited field averages. ¹²
Disinfection optimization	Move scheduled disinfections to times that target only idle machines; eliminate redundant cycles while staying within manufacturer instructions	ANZSN case study: 25% fewer disinfections , >8,000 fewer cycles/year , >160,000 L water saved , 6.4 MWh power saved , 720 L citric acid solution avoided , and \$2,400 product savings. Home HD example: 30% fewer disinfections for thrice-weekly patients. ¹³	High scalability because it uses existing equipment and manufacturer guidance. Patient-safety risk is manageable only if schedule changes remain within machine-specific requirements; ANZSN explicitly states there is no patient-safety benefit from redundant disinfections. ¹³

Practice	What is being done	Measured or reported effect	Safety and evidence status
Heat recovery	Install machine-level or central heat exchangers to transfer heat from spent dialysate/ wastewater to incoming water	East Kent retrofits saved 0.86 kWh/treatment and projected 27,905 kWh/year across 52 machines. Newcastle home HD observed 0.625 kWh/treatment in summer. B. Braun reports up to 17% annual savings from central heat recovery. A 2026 model validated experimentally projected >15 kWh/session and ~63,200 kWh/year for a four-station clinic. ¹⁴	Moderate-to-strong evidence for engineering feasibility and energy benefit. Real-world data exist, but patient-centre adoption is still limited compared with reject-water reuse. ¹⁵
Closed-loop or near-closed-loop reuse	Retreatment of reject water and reintegration into wider facility supply; blending with rainwater or other sources; in research, purification of spent dialysate	Queen Elizabeth Hospital in Adelaide filtered, rechlorinated, and blended RO reject with rainwater before return to site supply, saving 1.5–2 ML/year ; capex for the broader water-recovery plan was estimated at A\$500,000–A\$1 million . ¹⁶	Technically possible, but capital- and governance-intensive. Evidence is sparse and site-specific. Safety assurance is much more demanding than simple non-potable reuse. ¹⁷
Sorbent dialysis and portable low-water systems	Regenerate a small dialysate volume using sorbents rather than single-pass large-volume water	The REDY system used about 6 L untreated water per dialysis versus 400–600 L/treatment for conventional single-pass systems. NeoKidney reports 4.5–5 L/session , ~10 kg weight, and no separate water-treatment system. ¹⁸	Resource potential is very large, but implementation maturity is still limited. NeoKidney states that CE/FDA approval is pending and the device is not yet commercially available. ¹⁹
Batch/low-flow home systems	Use batch preparation and lower dialysate-flow prescriptions to reduce source-water demand relative to conventional home RO setups	Fresenius' NxStage materials state that PureFlow prepares 40/50/60 L batches, can support up to three treatments , converts source water to product water on a 1:1 basis, and "requires less water than an RO machine to produce the same amount of product water." NxStage guidance also notes that slower dialysate flow allows efficient clearance with less fluid. ²⁰	Moderate maturity in home HD, but this is <i>water-reduction</i> rather than a reuse technology. Savings depend heavily on prescription, frequency, and local home setup. ²¹

What the current evidence says

The evidence hierarchy is clear. **Reject-water reuse and disinfection optimization are today's highest-confidence measures** because they avoid patient-contact water pathways, use familiar engineering, and already have repeated service-level case studies with water, cost, and sometimes carbon metrics. **Heat recovery** is a credible next tier: good engineering logic, some implementation evidence, but still fewer large-scale replications. **Portable sorbent systems and spent-dialysate regeneration** offer the largest theoretical water savings, but the present evidence base is still too early for broad implementation recommendations outside controlled pilots. ²²

Operational practices

Operational practice is not a side issue in dialysis sustainability; it is often the reason identical equipment performs very differently across sites. The most consistent operational findings in the retrieved literature are that **higher utilization, fewer redundant cleaning cycles, formal training and competency systems, continuous monitoring, and good maintenance discipline** all reduce resource intensity without lowering the patient-safety bar. ²³

Operational lever	What the source says	Reported outcome
Scheduling and asset utilization	In the Spanish 20-centre study, centres opening daily had lower resource intensity than centres opening only three days/week . This is best interpreted as a load-factor effect: central plant fixed consumption is diluted over more treatments when equipment is used more continuously. ²⁴	430.20 L/HD and 10.82 kWh/HD in daily-opening centres versus 629.81 L/HD and 16.79 kWh/HD in centres opening three days/week. ²⁴
Patient grouping and fuller shifts	No retrieved trial explicitly randomized patient grouping, but the same multicentre data imply that grouping treatments to improve daily utilization should reduce resource intensity per session. This is therefore an <i>inference</i> rather than a standalone tested intervention. ²⁴	Quantitative effect not separately reported; inferred from daily-use advantage.
Staff training and sustainability engagement	ERA/SFNDT states that staff involvement and training are “fundamental”; ACI NSW requires educated, trained, and accredited operators, multidisciplinary committee review, annual audits, and documented policies. ²⁵	Direct savings unspecified, but these are enabling conditions for safe implementation.
Water testing, trending, and accreditation	ACI NSW requires routine chlorine testing, trending of results, operator accreditation, maintenance records, and multidisciplinary review. It notes that >95% rejection is an adequate RO-performance indicator and that all interventions must be logged. ²⁶	Direct savings unspecified; risk reduction and early fault detection are the main operational benefits.

Operational lever	What the source says	Reported outcome
Disinfection schedule redesign	ANZSN shows that moving automated disinfection so that only idle machines run the extra cycle can materially reduce water, power, and chemical use. ¹³	25% fewer disinfections, >160,000 L water and 6.4 MWh power saved annually in one service. ¹³
Maintenance discipline	Bendigo's reuse project required infection-prevention involvement, risk-management plans, dual-pipe guidance, and routine checks on pumps, bearings, pipelines, and tanks; ACI NSW requires log books and recorded interventions for all water-plant work. ²⁷	Direct savings unspecified; the point is reliability and safety, not merely conservation.
Design-in versus retrofit	Ashford's reuse system cost £2,500 because water recovery was designed into a new build; Canterbury's retrofit cost £15,000 . Bristol also reported that retrofitting a greywater system was prohibitively expensive, leading the team to redirect reject water to steam production instead. ²⁸	Strong evidence that greenfield designs improve economics.

A second operational lesson is that **supply-demand matching matters as much as supply volume**. The Australian handbook emphasizes that RO reject water is generated predictably during the working day and often matches toilet-flushing demand well, whereas irrigation is seasonal and may need storage or may not actually substitute potable demand. It therefore recommends evaluating projects by the percentage of reject water *used*, not merely the amount generated. ²⁹

A third lesson is that **monitoring should be institutionalized, not improvised**. The Spanish multicentre study concludes that annual monitoring of environmental key performance indicators is necessary to detect variation and guide action plans. Fresenius reports automated water meters in water-stressed U.S. clinics to improve treatment-level visibility and intervention targeting. These are signs that dialysis sustainability is moving from anecdotal “green projects” toward operational management. ³⁰

Policy drivers and stakeholder landscape

Dialysis sustainability is shaped by two different policy layers. The first is the **traditional water-quality and infection-control layer**, which is already mature and non-negotiable. ACI NSW requires written policies, trained and accredited operators, routine chlorine and post-RO testing, trending, audits, committee review, and maintenance logs; it also references ISO 13959 for contaminant control. These rules are the reason why sustainability measures that remain **outside the patient-contact water loop** scale more easily than measures that try to re-enter it. ²⁶

The second layer is the **climate, reporting, procurement, and resilience layer**, which is newer. WHO's guidance for climate-resilient and environmentally sustainable health facilities explicitly frames water, energy, procurement, staff training, and monitoring as central operational tasks. Fresenius Medical Care's 2025 sustainability statement is prepared under the EU CSRD/ESRS reporting regime. DaVita cites the U.S. Health Sector Climate Pledge. ISN's GREEN-K initiative is explicitly pushing a procurement

agenda so that manufacturers deliver more climate-resilient kidney care systems, and the ERA/SFNDT guide is trying to translate this into dialysis-centre action plans. ³¹

A practical implication follows from this layered policy structure: **dialysis-specific safety regulation is already detailed, while dialysis-specific sustainability incentives are still mostly indirect.** The clearest explicit incentives identified in the retrieved primary material were project grants and invest-to-save mechanisms rather than direct reimbursement: the Victorian **Greening Our Hospitals** grants, **Smart Water Fund**, **Stormwater and Urban Water Conservation Fund**, and UK-style internal business-case funding and interest-free efficiency loans such as **Salix Finance**. ³²

Key stakeholders and what they are doing

Stakeholder	Specific program or project	What they are doing	Reported result
East Kent Hospitals University NHS Foundation Trust and Centre for Sustainable Healthcare	Canterbury/Ashford reject-water recovery; Kent heat-exchanger retrofit	Reuse RO reject for toilets and laundry; retrofit heat exchangers to existing machines	Canterbury: 800 L/h reject-water recovery, £7,500/year savings; Ashford new-build reuse cost £2,500 ; 52-machine heat-exchanger retrofit projected 27,905 kWh/year and £2,498.60/year savings. ³³
University Hospitals Bristol NHS Foundation Trust	Bristol Royal Hospital for Children reject water to steam	Redirect reject water to boiler-house steam generation after greywater retrofit proved too costly	13,140 m³/year water saving, >£15,000/year , 13,836 kg CO₂e/year , £45,000 capex. ³⁴
Barwon Health, Bendigo Health, East Wimmera, Queen Elizabeth Hospital	Australian public-service reuse projects	Large-scale reuse for sterilizers, toilets, cooling towers, irrigation, and blended retreated supply	Barwon measured about 100 kL/week across facilities and homes; Bendigo 1.7 ML/year ; Donald 1,000 L/week ; QEH 1.5–2 ML/year . ³⁵
ANZSN	Dialysis machine disinfection scheduling guide	Standardize lower-waste disinfection scheduling without violating manufacturer requirements	One case study saved >160,000 L water and 6.4 MWh power annually. ¹³
Fresenius Medical Care	Water-stress assessments, newer RO systems, water metering, heat disinfection	Corporate water-risk planning; newer systems with higher recovery; site metering in water-stressed clinics	Company reports 34.8 million m³ withdrawal and 106.9 million m³ water reused/recycled in 2025, with method caveats; newer RO systems reported with up to 85% recovery . ³⁶

Stakeholder	Specific program or project	What they are doing	Reported result
DaVita	Water-efficiency projects and broader ESG program	Corporate water-reduction program across outpatient centres; renewable-energy and net-zero-centre work	>75 million gallons saved in 2024 and 240 million gallons since 2021 ; also opened the first U.S. all-electric net-zero dialysis centre. ³⁷
B. Braun	AQUAboss RO and central heat recovery	High-recovery RO systems and central wastewater heat recovery concept	AQUAboss claims up to 85% recovery; central heat-recovery concept reports up to 17% annual energy savings. ³⁸
NextKidney and sorbent developers	NeoKidney portable HD; legacy REDY concept	Low-water sorbent regeneration to avoid large home water-treatment systems	REDY: ~6 L/treatment ; NeoKidney: 4.5–5 L/session , not yet for sale, approvals pending. ³⁹
SFNDT, ERA, ISN GREEN-K, KDIGO	Green dialysis guides and procurement agenda	Professional guidance, education, and a push to influence procurement and innovation	ERA/SFNDT identifies staff training as fundamental; ISN GREEN-K focuses on sustainable procurement; KDIGO's 2026 conference report elevates green dialysis to international policy discussion. ⁴⁰

Quantified outcomes, geography, and scalability

The measured outcomes retrieved in this review are concentrated in a few geographies. **The UK** provides some of the clearest business-case case studies for reject-water reuse and heat exchangers. **Australia** contributes much of the best implementation detail on RO-reject reuse, reuse handbooks, water-safety governance, and disinfection optimization. **Spain** contributes the best multicentre observational data on how plant choices and operations change environmental intensity. **The Netherlands, Singapore, Morocco, and the U.S.** are more visible in emerging-device and system-level innovation than in fully mature scaled deployment. ⁴¹

Measured outcomes table

Site or study	Practice	Water outcome	Energy / carbon outcome	Cost outcome	Safety / clinical note
Canterbury Dialysis Unit, UK	RO reject reused for hospital toilets	800 L/h recovered	Unspecified	£15,000 capex; £7,500/year savings; system operated for >10 years	Source states reject water poses no infection risk because it does not contact the patient. ⁴²
Ashford Dialysis Unit, UK	New-build reject-water reuse to laundry	Water quantity unspecified	Unspecified	£2,500 implementation cost	Demonstrates design-in advantage; safety outcomes unspecified. ⁴²
Bristol Royal Hospital for Children, UK	Reject water diverted to boiler house for steam	13,140 m³/year	13,836 kg CO₂e/year	>£15,000/year savings; £45,000 capex; ~67% five-year ROI calculation reported	Non-patient-contact reuse; no infection outcome reported. ³⁴
East Kent renal service, UK	Machine heat-exchanger retrofit	Water unspecified	0.86 kWh/treatment; 27,905 kWh/year across 52 machines; 16.46 t CO₂e/year	£48.05/machine/year and £2,498.60/year across 52 machines; device payback <4 years	No adverse patient-safety signal reported; intervention affects heating efficiency, not water purity. ⁴³
Newcastle home HD, UK	Home-machine heat-exchanger retrofit	Water unspecified	0.625 kWh/treatment observed in summer; 97.5 kWh/year/machine at 3 sessions/week; 1.207 t CO₂e/year across 21 machines	~£15.60/year/machine at the quoted tariff; £327.60/year across 21 machines	Savings likely larger in winter because colder incoming water increases baseline energy use. ⁴⁴

Site or study	Practice	Water outcome	Energy / carbon outcome	Cost outcome	Safety / clinical note
Barwon Health, Australia	Reuse cascade for sterilizing water, cooling towers, toilets, irrigation	>20 kL/ week at one site (~1.1 ML/ year) reported; broader service measured ~100 kL/ week	Unspecified	~\$36,000 annual water value cited for one site	Illustrates cascading reuse and multi-end-use operation; direct infection outcomes unspecified. ⁴⁵
Bendigo Health, Australia	RO reject to toilet flushing	~1.7 ML/ year potable substitution	Unspecified	Funding via Stormwater and Urban Water Conservation Fund; direct operating savings unspecified	Formal infection-prevention, risk-management, and dual-pipe guidance were part of implementation. ⁴⁶
Donald, Australia	Small-unit RO reject to irrigation	~1,000 L/ week	Unspecified	A\$4,110 system cost, funded by Greening Our Hospitals	Good example of small rural-unit feasibility; clinical outcomes unspecified. ⁴⁶
Queen Elizabeth Hospital, Australia	Filtered/ rechlorinated reject water blended with rainwater and returned to site supply	1.5–2 ML/ year	Unspecified	Broader water-recovery plan estimated at A\$500,000–A\$1 million	High-governance, high-capex model; much more complex than non-potable reuse. ¹⁶
ANZSN case study, Australia	Disinfection scheduling optimization	>160,000 L/ year	6.4 MWh/ year	\$2,400 product savings; 720 L citric acid avoided	ANZSN explicitly says redundant disinfections provide no patient-safety benefit . ¹³

Site or study	Practice	Water outcome	Energy / carbon outcome	Cost outcome	Safety / clinical note
Multicentre Spain study, 20 HD centres	Facility and operational optimization over five years	Mean 476.5 L/HD across all centres; 430.20 L/HD in daily-opening centres vs 629.81 L/HD in centres open three times weekly	Mean 12.6 kWh/HD overall; 10.82 kWh/HD daily vs 16.79 kWh/HD three times weekly	Site-specific cost data unspecified	Corrective actions reduced energy 6–30%, plastic waste 62–65%, and chemical waste 63%. ²⁴
Spanish multicentre extrapolation	Annualized burden per patient on in-centre HD	74,326.2 L/patient/year	1,962.5 kWh/patient/year	Unspecified	Waste burden also quantified at 376.4 kg/patient/year. ²⁴
Modern versus older Australian RO systems	RO-plant modernization	357 L/treatment versus 548 L/treatment	3.1 kWh/treatment versus 7.2 kWh/treatment	Unspecified	Independent comparison supports plant renewal as a major sustainability lever. ⁴⁷
AWAK PD pilot, Singapore	Sorbent/regenerative dialysis platform	Water use not reported in the trial abstract retrieved here	Not reported	Unspecified	15 patients; no serious adverse events; abdominal pain/discomfort in 60%; significant reductions in serum urea, creatinine, phosphate, and β 2-microglobulin. ⁴⁸

Geographic distribution and scalability

The **highest-scalability practices** are the ones that require the smallest change to the regulated patient-contact pathway: disinfection scheduling, metering, staff training, maintenance, and

specification of better RO plants. Those can often be rolled out service-wide with modest capital spending and little regulatory novelty. The next tier is **RO reject-water reuse**, which scales well where a stable nearby demand exists—especially toilets, laundry, sterilizers, and boilers. Scalability drops when storage, pumping, or long-distance conveyance is needed, or when local non-potable plumbing standards become stringent. ⁴⁹

The **lowest current scaling readiness** belongs to patient-contact closed-loop concepts. Even where the technical promise is strong, the evidence base is still too concentrated in engineering studies, prototype brochures, conference abstracts, and small pilots. That does not make them unimportant; it means they are better treated today as **strategic innovation tracks** rather than as default infrastructure choices for broad service rollouts. ⁵⁰

Barriers and best-practice recommendations

Main barriers and risks

Barrier	Why it matters	Evidence
Retrofit cost and building constraints	Reuse can be cheap when designed into a new unit but expensive when storage, pumps, pipework, and approvals are retrofitted into an existing hospital	Ashford cost £2,500 as a new-build feature versus £15,000 at Canterbury; Bristol explicitly said a greywater retrofit was prohibitively expensive, so it chose the boiler-house route instead. ²⁸
Demand-supply mismatch	Reject water is useful only if there is a nearby, steady demand; irrigation often has poor temporal alignment	The Australian handbook notes that RO reject is predictable in daytime and well matched to toilet demand, whereas irrigation is seasonal and may not substitute potable demand. ²⁹
Water-quality and safety governance	Any change in water routing requires fit-for-purpose testing, documented procedures, and operator competency; patient-contact reuse is especially demanding	ACI NSW requires testing, trending, accredited operators, multidisciplinary review, and maintenance logs; the handbook recommends direct measurement where consequences of poor water are significant. ⁵¹
Aging plants and legacy settings	Old plants tend to have poorer recovery and higher water/energy use; bad settings can wash out sustainability gains	ERA/SFNDT cites a French survey in which 33% of responding units had water-treatment plants older than 15 years with poor environmental performance; independent data show large differences between old and modern RO systems. ⁵²
Limited clinician and organizational engagement	Sustainability programs stall without champions, data, and staff buy-in	A multinational survey summarized in <i>Blood Purification</i> found only 22% of participants involved in sustainable kidney care initiatives and 14% in climate-change-and-kidney-health initiatives. ⁵³

Barrier	Why it matters	Evidence
Economic signal problems	Water savings do not always translate proportionally into bill savings because fixed service charges can rise or dominate	Barwon noted that later water-authority service-charge increases meant dollar savings no longer tracked liters saved exactly. ⁵⁴
Immature evidence for emerging low-water systems	Innovative options can promise dramatic water savings but still lack mature clinical, regulatory, or commercial proof	NeoKidney is still pending regulatory approval; AWAK required device alterations during the pilot and further enhancements to reduce adverse events; forward-osmosis regeneration remains largely research-stage. ⁵⁵

Best-practice recommendations

Start with measurement, not ideology. Every unit should baseline at least four indicators: water per treatment, RO reject volume, energy per treatment, and number of disinfection cycles. The Spanish multicentre study and Fresenius' meter rollout both point in the same direction: the units that can see their water and energy use are the ones that can manage it. ³⁰

Optimize before you invent. The fastest wins usually come from adjusting schedules, removing redundant disinfection, renewing aging RO plants, optimizing regeneration frequencies, and fixing leaks or overuse. These are lower-risk measures than closed-loop recycling and often deliver immediate water and energy reductions. ⁵⁶

Reuse only the safest streams first. The best near-term default is to prioritize **RO reject water** and other **non-patient-contact** streams. The economic and safety evidence is strongest there. Reserve patient-contact closed-loop approaches for research or carefully governed pilots until more clinical and regulatory evidence accumulates. ⁵⁷

Design sustainability into new builds. New unit design should explicitly include reject-water reuse, storage, meters, and heat recovery opportunities where feasible. The cost gap between Ashford and Canterbury is a powerful reminder that sustainability is much cheaper when planned before the walls are closed. ⁴²

Match reuse to stable demand. Toilet flushing, sterilizer/cooling water, boilers, and laundry typically beat irrigation because they provide predictable year-round demand and avoid overbuilding storage. Project appraisal should focus on *utilization rate* of recovered water, not on gross reject-water generation. ⁵⁸

Institutionalize competence and governance. A multidisciplinary water committee, operator accreditation, annual audits, and a maintenance logbook are not bureaucratic extras; they are the controls that make sustainability changes safe enough to scale. ²⁶

Use procurement strategically. When renewing plants or machines, require evidence on recovery rate, heat-disinfection capability, telemetry/metering, and compatibility with heat-recovery or reuse architectures. This aligns with ISN GREEN-K's explicit procurement strategy and with the direction of European and corporate sustainability reporting. ⁵⁹

Open questions and assumptions

Direct **patient safety, infection, and hospitalization outcomes** attributable to sustainability measures remain underreported. Most retrieved sources demonstrate safety through water-quality control protocols or by keeping reuse outside the patient-contact pathway, rather than through comparative infection studies. That is a meaningful evidence gap. ⁶⁰

Evidence for **closed-loop spent-dialysate recycling, forward osmosis, and portable sorbent hemodialysis** is still not robust enough to support broad deployment advice on its own. The resource potential is substantial, but the field still needs more peer-reviewed clinical data, better published lifecycle assessments, and transparent cost comparisons at scale. ⁶¹

A final practical limitation is that some corporate and manufacturer metrics are **company-reported** and sometimes **extrapolated**, not independently audited at the level of each dialysis session or each site. They are useful directional signals, but they should not be treated as identical in evidentiary weight to a fully published multicentre observational study. ⁶²

¹ ⁶ ⁸ ⁹ ¹⁰ ²² ²⁸ ³³ ⁴¹ ⁴² ⁵⁷ <https://production.networks.sustainablehealthcare.org.uk/resources/case-study-reuse-ro-reject-water-two-haemodialysis-units>

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¹³ ⁴⁹ ⁵⁶ <https://d122d2wjqead0l.cloudfront.net/AP/Int/anzsn/uploads/Green%20Neph/ESC%20Optimising%20Disinfection%20of%20Dialysis%20machines.pdf>

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